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7 Top-Down Scan Insertion

Learning Objectives

After completing this lab, you should be able to:

- Use the `preview_dft` command to evaluate different scan architectures and iterate by revising scan specifications until you produce balanced top-level scan chains.
- Use the `preview_dft` command to confirm your scan specification for desired scan signals was applied correctly.

Lab Duration:
45 minutes

Lab 7

Overview

During this lab you will:

1. Preview the results of different scan specifications before implementing one that yields well-balanced top-level chains.
2. Perform the Mapped DFT flow for a test-ready design through scan insertion and test coverage estimation.

Unmapped Flow:

1. Read the RTL design into DFT.
2. Create and save a test protocol; verify that the design and the test protocol are compatible.
3. Compile and save the gate-level design; exit.

Mapped Flow:

4. Read the gate-level design and test protocol into DFTC.
5. Specify scan constraints and preview the scan architecture that will result from applying them.
6. Insert the scan chain.
7. Hand off the design and related files for use by downstream tools; exit.

File Locations

All files for this lab except for designs and libraries are located in the directory **lab7_topdown**.

Directory Structure

lab7_topdown	Current working directory
analyzed	Intermediate acs_read_hdl files
logs	Session log files
unmapped	Unmapped protocol
mapped	Gate-level netlist
mapped_scan	Gate-level scan design netlist
reports	DFTC reports
tmax	Files for downstream tools
scripts	Constraint and run scripts
../ref/rtl/ORCA/vhdl	Design files
../ref/db	Library files

Instructions

Your goal is to complete and run the first three steps of the Mapped Flow portion of the DFT Insertion Flow. The three you will modify are for reading the test-ready design and test protocol, previewing the scan architecture, inserting scan chains and finally, estimating test coverage.

The three scripts:

```
Unmapped Flow
  1read_design.tcl
  2create_test_protocol.tcl
  3compile_and_save.tcl
Mapped Flow
  4read_gates_and_protocol.tcl
  5preview_dft.tcl
  6insert_dft.tcl
  7handoff.tcl
```

Task 1. Read Gate-level Design and its Test Protocol

1. Invoke DFTC.

```
cd lab7_topdown
dc_shell | tee logs/lab7.log
```

2. Examine and execute the script that performs step 4 of the DFT flow: reading in the gate-level design and the saved protocol.

```
exec cat scripts/4read_gate_and_protocol.tcl

s scripts/4read_gate_and_protocol.tcl
```

连续做lab6, lab7、
⚠要退出上一个design的文件夹载入. 再进入lab7.运行source. 否则会有文问

```
dc_shell> report_scan_state

****************************************
Report : test
        -state
Design : ORCA
Version: G-2012.06
Date   : Tue Sep 11 20:12:34 2018
****************************************

Scan state                    : scan cells replaced with loops

1
```


3. Check the scan state of the current design using this command:

```
report_scan_state
```

Question 1. If the Unmapped Flow ended with a `compile -scan` command, what should the scan state of the design you just read in be? Is that the design's current scan state?

scan cells replaced with loops.
yes, that is.

At the end of the **4read_gate_and_protocol.tcl**, a DFT check was performed.

4. Examine the last several lines in your transcript to review the DFT check results for the test-ready design.

Question 2. What percentage of the total number of flip-flops are valid scan cells?

```
SEQUENTIAL CELLS WITH VIOLATIONS
      *  32 cells have test design rule violations
SEQUENTIAL CELLS WITHOUT VIOLATIONS
      *2926 cells are valid scan cells
```

```
2926 ÷ 2958 =
0.98918187964841108857336037863421

2926 + 32 =
2,958
```

Question 3. Considering just that percentage and no other factors, what range of estimated test coverage would you predict for this design?

接近上面的计算值吧

5. Enable enhanced DRC reporting and re-run `dft_drc` # 牛逼模式 ON！

```
set test_disable_enhanced_dft_drc_reporting FALSE
dft_drc
```

Question 4. Does the percentage reported by the enhanced `dft_drc` reporting match your expectations?

符合。（精度真低）

```
-----------------------------------------------------------------------------------
Sequential Cell Report:        Sequential        Core              Core
                               Cells             Segments          Segment Cells
-----------------------------------------------------------------------------------
Sequential Elements Detected:   2958                0                   0
Clock Gating Cells          :   0
Synchronizing Cells         :   0
Non-Scan Elements           :   0
Excluded Scan Elements      :   0                   0                   0
Violated Scan Elements      :   32                  0                   0
(Traced) Scan Elements      :   2926    ( 98.9%)    0       (  0.0%)    0       (  0.0%)
-----------------------------------------------------------------------------------
```

Note: Step 4 of the DFT flow reads the gate-level design and the test protocol. The dft_drc command used the saved test protocol for DFT design rule checking. The test protocol is one set of information and the scan specifications that may have been used to originally create the test protocol is another set of information.

6. Check this by reporting which dft signals are currently defined.

```
report_dft_signal
```

显示signal列表

```
========================================
TEST MODE: all_dft
VIEW     : Existing DFT
========================================
Port                 SignalType           Active     Hookup     Timing     Usage
----------           ----------           ------     ------     ------     -----
test_mode            Constant             1          -
scan_en              ScanEnable           1          -
sys_clk              ScanMasterClock      1          -          P 100.0 R 45.0 F 55.0
sys_clk              MasterClock          1          -          P 100.0 R 45.0 F 55.0
sdr_clk              ScanMasterClock      1          -          P 100.0 R 45.0 F 55.0
sdr_clk              MasterClock          1          -          P 100.0 R 45.0 F 55.0
pclk                 ScanMasterClock      1          -          P 100.0 R 45.0 F 55.0
pclk                 MasterClock          1          -          P 100.0 R 45.0 F 55.0
prst_n               Reset                0          -          P 100.0 R 55.0 F 45.0
```

Lab 7

① Scan Clock 和 Master Clo

Question 5. How many ScanClocks and Resets are reported?

3个scanclocks (sys_clk,sdr_clk,pclk);
1个resets(prst_n)

Task 2. Specify and Preview Scan Architectures

1. Display the default scan architecture that will be implemented for this design.

P7-12

```
preview_dft
```

② 执行有问题的. Test Point Plan Report 我

```
****************************************
Preview_dft report
For     : 'Insert_dft' command
Design : ORCA
Version: G-2012.06
Date   : Tue Sep 11 20:48:31 2018
****************************************

Number of chains: 5
Scan methodology: full_scan
Scan style: multiplexed_flip_flop
Clock domain: no_mix
Scan enable: scan_en (no hookup pin)


Scan chain '1' (test_si1 --> test_so1) contains 1074 cells


Scan chain '2' (test_si2 --> test_so2) contains 620 cells


Scan chain '3' (test_si3 --> test_so3) contains 719 cells


Scan chain '4' (test_si4 --> test_so4) contains 1 cell


Scan chain '5' (test_si5 --> test_so5) contains 512 cells



************ Test Point Plan Report ************
Total number of test points  : 0
Number of Autofix test points: 0
Number of Wrapper test points: 0
Number of test modes         : 0
Number of test point enables : 0
Number of data sources       : 0
Number of data sinks         : 0
**************************************************
```

Question 6. How many scan chains will be inserted?
What are the names of the preview scan chains?
Will the chains be well-balanced?

5.
'1','2','3','4','5'.
1074:1, extreme unbalanced.

```
****************************************
Report : Scan configuration
Design : ORCA
Version: G-2012.06
Date   : Tue Sep 11 21:10:36 2018
****************************************

========================================
TEST MODE: all_dft
VIEW     : Specification
========================================
Chain count:                         Undefined
Scan Style:                          Multiplexed flip-flop
Maximum scan chain length:           Undefined
Exact scan chain length:             Undefined
Physical Partitioning:               Horizontal
Replace:                             True
Preserve multibit segments:          True
Clock mixing:                        No mix
Internal clocks:                     none
Add lockup:                          True
Lockup type:                         latch
Insert terminal lockup:              False
Create dedicated scan out ports:     False
Shared scan in:                      0
Bidirectional mode:                  No bidirectional type
Internal Clock Mixing:               False
Test Clocks by System Clocks:        False
Hierarchical Isolation:              False
Multiple Scan Enable:                Disable
Pipeline Scan Enable:                Disable
Voltage Mixing:                      False
Identify Shift Register:             False
Power Domain Mixing:                 False
Reuse MV Isolation Cells:            True
Multi LSSD:                          Disable
```

Question 7. Will scan insertion use existing functional pins as the scan signals (scan in, scan out for each scan chain) or will it create new ones (test_si, test_so ports)?

③ 在哪里看

it will create new (have not spec scan signals)

2. Display more scan architecture detail about the test clocks used for each scan chain.

```
preview_dft -show scan_clocks
```

Note: The ORCA design has only 3 clocks in test mode: pclk, sys_clk and sdr_clk.

Question 8. Why are five scan chains previewed by default?

五条链 五个时钟域

有5个时钟域，一条扫描链用一个。

```
Number of chains: 5
Scan methodology: full_scan
Scan style: multiplexed_flip_flop
Clock domain: no_mix
Scan enable: scan_en (no hookup pin)

  (l) shows cell scan-out drives a lockup latch
  (s) shows cell is a scan segment
  (w) shows cell scan-out drives a wire


Scan chain '1' (test_si1 --> test_so1) contains 1074 cells:

  I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_0_ (pclk, 45.0, rising)
     ...
  I_ORCA_TOP/I_RESET_BLOCK/prst_ff_reg


Scan chain '2' (test_si2 --> test_so2) contains 620 cells:

  I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_ff_reg (sdr_clk, 45.0, rising)
     ...
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/this_addr_g_int_reg_6_


Scan chain '3' (test_si3 --> test_so3) contains 719 cells:

  I_ORCA_TOP/I_BLENDER/rem_blue_reg (sys_clk, 45.0, rising)
     ...
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/this_addr_g_int_reg_6_1


Scan chain '4' (test_si4 --> test_so4) contains 1 cell:

  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg (sys_clk, 55.0, falling)


Scan chain '5' (test_si5 --> test_so5) contains 512 cells:

  I_ORCA_TOP/I_SDRAM_IF/DQ_out_1_reg_0_ (sdr_clk, 55.0, falling)
     ...
  I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__15_
```

3. Use another `preview_dft` option and compare the content of the former and current `preview_dft` output.

```
preview_dft -show scan_summary
```

Question 9. Is there any information in the scan_clocks preview that is not also reported in the scan_summary preview?
When might you want to use the format of the scan_summary preview rather than the first?

```
Number of chains: 5
Scan methodology: full_scan
Scan style: multiplexed_flip_flop
Clock domain: no_mix
Scan enable: scan_en (no hookup pin)

Chain      Scan Ports (si --> so)        # of Cells   Inst/Chain                  Clock (port, time, edge)
-----      ----------------------        ----------   ----------                  ------------------------
S 1        test_si1 -->  test_so1              1074   I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_0_
                            (pclk, 45.0, rising)
S 2        test_si2 -->  test_so2               620   I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_ff_reg
                            (sdr_clk, 45.0, rising)
S 3        test_si3 -->  test_so3               719   I_ORCA_TOP/I_BLENDER/rem_blue_reg
                            (sys_clk, 45.0, rising)
S 4        test_si4 -->  test_so4                 1   I_ORCA_TOP/I_BLENDER/latched_clk_en_reg
                            (sys_clk, 55.0, falling)
S 5        test_si5 -->  test_so5               512   I_ORCA_TOP/I_SDRAM_IF/DQ_out_1_reg_0_
                            (sdr_clk, 55.0, falling)
1
```

虽然列出了一张表格的形式，但是比上一种方法少反馈了最后一个不带clock的reg信息

scan chain过多，需要较简明的report的时候（？）

4. Try mixing the rising and falling sdr_clk scan flip-flops into the same chain.

```
set_scan_configuration -clock_mixing mix_edges
preview_dft -show scan_clocks
```

```
Number of chains: 3
Scan methodology: full_scan
Scan style: multiplexed_flip_flop
Clock domain: mix_edges
Scan enable: scan_en (no hookup pin)

  (l) shows cell scan-out drives a lockup latch
  (s) shows cell is a scan segment
  (w) shows cell scan-out drives a wire


Scan chain '1' (test_si1 --> test_so1) contains 1074 cells:

  I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_0_ (pclk, 45.0, rising)
     ...
  I_ORCA_TOP/I_RESET_BLOCK/prst_ff_reg


Scan chain '2' (test_si2 --> test_so2) contains 1132 cells:

  I_ORCA_TOP/I_SDRAM_IF/DQ_out_1_reg_0_ (sdr_clk, 55.0, falling)
     ...
  I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_ff_reg (sdr_clk, 45.0, rising)
     ...
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/this_addr_g_int_reg_6_


Scan chain '3' (test_si3 --> test_so3) contains 720 cells:

  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg (sys_clk, 55.0, falling)
  I_ORCA_TOP/I_BLENDER/rem_blue_reg (sys_clk, 45.0, rising)
     ...
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/this_addr_g_int_reg_6_1
```

Question 10. How many scan chains are previewed now and are they now balanced?

3.
相对较平衡了，但不够。

5. Allow mixing of different clock domain scan flip-flops into the same scan chain.

```
set_scan_configuration -clock_mixing mix_clocks
preview_dft -show scan_clocks
```

```
Number of chains: 1
Scan methodology: full_scan
Scan style: multiplexed_flip_flop
Clock domain: mix_clocks
Scan enable: scan_en (no hookup pin)

  (l) shows cell scan-out drives a lockup latch
  (s) shows cell is a scan segment
  (w) shows cell scan-out drives a wire


Scan chain '1' (test_si --> test_so) contains 2926 cells:

  I_ORCA_TOP/I_SDRAM_IF/DQ_out_1_reg_0_ (sdr_clk, 55.0, falling)
     ...
  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg (sys_clk, 55.0, falling)
  I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_0_ (pclk, 45.0, rising)
     ...
  I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_ff_reg (sdr_clk, 45.0, rising)
     ...
  I_ORCA_TOP/I_BLENDER/rem_blue_reg (sys_clk, 45.0, rising)
     ...
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/this_addr_g_int_reg_6_1
```

Question 11. How many scan chains are previewed now and what do you think the default `set_scan_configuration -chain_count` value is?

1.
6.

如这6条. 是否跨时钟域 是的!

☆ **Question 12.** What specifications would yield 6 balanced scan chains? 如何得6条

```
dc_shell> set_scan_configuration -chain_count 6
```

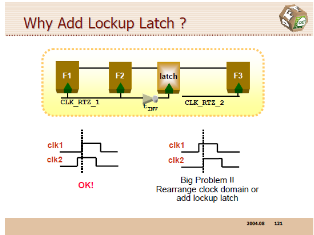
-chain_count *chain_count* 指定要构建的扫描链数。

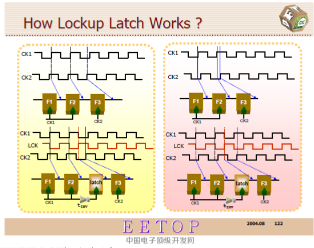
Specifies the number of scan chains to build. The tool attempts to meet this constraint. If unable to do so, the tool issues a warning. See the DESCRIPTION section for more information.

-clock_mixing no_mix | mix_edges | mix_clocks | mix_clocks_not_edges

Specifies whether the **insert_dft** command can include cells from different clock domains in the same scan chain. The following allowed values control the clocking of cells in scan chains to be inserted by the **insert_dft** command:

- **no_mix** (the default)
  Cells must be clocked by the same edge of the same clock.
- **mix_edges**
  Cells must be clocked by the same clock, but the clock edges can be different.
- **mix_clocks_not_edges**
  Cells must be clocked by the same clock edge, but the clocks can be different.
- **mix_clocks**
  Cells can be clocked by different clocks and different clock edges.






```
Scan chain '1' (test_si1 --> test_so1) contains 488 cells:

  I_ORCA_TOP/I_SDRAM_IF/DQ_out_1_reg_0_ (sdr_clk, 55.0, falling)
    ...
  I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__7_

Scan chain '2' (test_si2 --> test_so2) contains 488 cells:

  I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__8_ (sdr_clk, 55.0, falling)
    ...
  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg (sys_clk, 55.0, falling)
  I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_0_ (pclk, 45.0, rising)
    ...
  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_26__6_

Scan chain '3' (test_si3 --> test_so3) contains 488 cells:

  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_26__7_ (pclk, 45.0, rising)
    ...
  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_56__14_

Scan chain '4' (test_si4 --> test_so4) contains 488 cells:

  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_56__15_ (pclk, 45.0, rising)
    ...
  I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_ff_reg (sdr_clk, 45.0, rising)
    ...
  I_ORCA_TOP/I_SDRAM_IF/mega_shift_0_reg_20__11_

Scan chain '5' (test_si5 --> test_so5) contains 487 cells:

  I_ORCA_TOP/I_SDRAM_IF/mega_shift_0_reg_20__12_ (sdr_clk, 45.0, rising)
    ...
  I_ORCA_TOP/I_BLENDER/rem_blue_reg (sys_clk, 45.0, rising)
    ...
  I_ORCA_TOP/I_BLENDER/s4_op1_reg_4_

Scan chain '6' (test_si6 --> test_so6) contains 487 cells:
```

Apply those specifications (6 chains) and confirm using preview_dft.

```
set_scan_configuration    # specify your options
preview_dft -show scan_clocks
```

Question 13. What is the difference in scan chain length between the longest and shortest chains now?

Almost no difference

```
Scan chain '2' (test_si2 --> test_so2) contains 488 cells:

  I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__8_ (sdr_clk, 55.0, falling)
    ...
  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg (sys_clk, 55.0, falling)
  I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_0_ (pclk, 45.0, rising)
    ...
  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_26__6_

Scan chain '3' (test_si3 --> test_so3) contains 488 cells:

  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_26__7_ (pclk, 45.0, rising)
    ...
  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_56__14_

Scan chain '4' (test_si4 --> test_so4) contains 488 cells:

  I_ORCA_TOP/I_PCI_CORE/mega_shift_reg_56__15_ (pclk, 45.0, rising)
    ...
  I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_ff_reg (sdr_clk, 45.0, rising)
    ...
  I_ORCA_TOP/I_SDRAM_IF/mega_shift_0_reg_20__11_

Scan chain '5' (test_si5 --> test_so5) contains 487 cells:

  I_ORCA_TOP/I_SDRAM_IF/mega_shift_0_reg_20__12_ (sdr_clk, 45.0, rising)
    ...
  I_ORCA_TOP/I_BLENDER/rem_blue_reg (sys_clk, 45.0, rising)
    ...
  I_ORCA_TOP/I_BLENDER/s4_op1_reg_4_
```

Question 14. Will DFTC insert any lock-up latches into this scan design? Which chains will include them?

yes.
2,4,5.

Note: The functional pins of ORCA to use for scan are specified in the settings_insert_dft.tcl script.

7. Source that script and use `preview_dft` to answer the following questions.

```
s scripts/settings_insert_dft.tcl
preview_dft
```

Question 15. What are the names of the previewed scan chains now and what command defined their new names?
Why were the chains given explicit names?

```
Scan chain 'chain0' (pad[0] --> sd_A[0]) contains 488 cells
Scan chain 'chain1' (pad[1] --> sd_A[1]) contains 488 cells
Scan chain 'chain2' (pad[2] --> sd_A[2]) contains 488 cells
Scan chain 'chain3' (pad[3] --> sd_A[3]) contains 488 cells
Scan chain 'chain4' (pad[4] --> sd_A[4]) contains 487 cells
Scan chain 'chain5' (pad[5] --> sd_A[5]) contains 487 cells
```

单纯的数字前面多了‘chain’；set_scan_path...

```
set_scan_path chain$i -view spec -scan_data_in pad[$i] -scan_data_out sd_A[$i]
```

```
# Specify all scan ports
for {set i 0} {$i < 6} {incr i} {
```

Question 16. Which pins are used for scan in, scan out and scan enable?

```
set hookup_cell pad_iopad_$i
set_dft_signal -view spec -port pad[$i] -type ScanDataIn -hookup_pin $hookup_cell/CIN
set hookup_cell sdram_A_iopad_$i
set_dft_signal -view spec -port sd_A[$i] -type ScanDataOut -hookup_pin $hookup_cell/I
```

```
TEST MODE: Internal_scan
VIEW     : Specification
===============================
Port              SignalType
----------        ----------
pad[0]            ScanDataIn
sd_A[0]           ScanDataOut
pad[1]            ScanDataIn
sd_A[1]           ScanDataOut
pad[2]            ScanDataIn
sd_A[2]           ScanDataOut
pad[3]            ScanDataIn
sd_A[3]           ScanDataOut
pad[4]            ScanDataIn
sd_A[4]           ScanDataOut
pad[5]            ScanDataIn
sd_A[5]           ScanDataOut
```

```
TEST MODE: Internal_scan
VIEW     : Existing DFT
===============================
Port              SignalType
----------        ----------
test_mode         Constant
scan_en           ScanEnable
sys_clk           ScanMasterClock
sys_clk           MasterClock
sdr_clk           ScanMasterClock
sdr_clk           MasterClock
pclk              ScanMasterClock
pclk              MasterClock
prst_n            Reset
```

Task 3. Insert Scan Chains and Estimate Test Coverage

One of the commands in the settings_insert_dft.tcl script avoids all the subdesigns being renamed during the scan insertion process.

?没有这句啊

Question 17. What is the name of that command?

`set_dft_insertion_configuration` ....

```
-preserve_design_name true | false
        Specifies whether to preserve the design name when writing  from
        the tool to the database during DFT insertion.  Valid values are
        true to preserve the design name, and false to  permit  renaming
        the design.

        The default value is false.
```

Note: Use the man pages if needed to find the option to this command that disables all the gate-level optimization that occur by default during the last phases of the `insert_dft` process.

Question 18. Which option to this command will disable gate-level optimization?

....

```
-synthesis_optimization none | all
        Specifies whether to perform optimizations during DFT insertion.

        In  wireload mode, valid values are all to turn on all optimiza-
        tions, and none to turn off all optimizations. The default value
        is all.

        In topographical mode, the only allowed value is none; it cannot
        be changed.
```

1. Apply those two options whose effects will be to speed-up the runtime of the subsequent `insert_dft` and to limit the number of changes to the existing design.

insert_dft.

```
set_dft_insertion_configuration . . .
```

Perform scan insertion using the script provided.

```
s scripts/6insert_dft.tcl
```

```
    Uncollapsed Stuck Fault Summary Report
-----------------------------------------------
fault class                     code   #faults
------------------------------  ----  ---------
Detected                         DT      68988
Possibly detected                PT         67
Undetectable                     UD       1233
ATPG untestable                  AU       3274
Not detected                     ND        108
-----------------------------------------------
total faults                             73670
test coverage                           95.28%
-----------------------------------------------
```

逻辑—Fault
时序—scan chain
通过逻辑的值传到寄存器
然后输出
②如何提高 coverage.
是dft_drc violation
导致coverage低.
降低violation 则coverage
提高

Question 19. What is the estimated test coverage for the ORCA scan design?

95.28% ....

Note: Actual ATPG runs in TetraMAX obtained about 99% test coverage for this design. Doing so though required using functional RAM models and both TetraMAX's Fast Sequential and Full Sequential engines. Using DFT techniques to avoid the S30 violation, for example, allows that very high coverage much easier to obtain by merely using the Fast Sequential engine.

Question 20. How does this estimate compare to your earlier prediction?

95.28%着实比99%低太多了。。 ....

Note: When obtaining the estimated fault coverage for a scan inserted design, `dft_drc` performs Post-DFT checking and reports the DFT violations using the same ones that TetraMAX does.

```
46 DRC VIOLATIONS WHICH CAN AFFECT ATPG COVERAGE
    1 Clock connected to primary output violation (C17)
    1 Multiply clocked scan chain violation (S22)
   44 Bidi bus driver enable affected by scan cell violations (Z9)
```

Lab 7

Note: You can obtain detailed information for both the D* PRE-DFT violations and the {C,S,Z,X}* Post-DFT violations using the man command in DFTC.

C17 S22 Z9
clock scanchain tristate

3. Scroll back from the coverage report and answer the following questions.

pre-DFT 和 post-DFT 流程上差在哪
是差在 insert-dft 么. √

Question 21. The S category of violations cannot be reported during Pre-DFT checks, why?

..S类的violation与scan chain有关（详见tmax_rules.pdf），而在pre-dft时候，还未插入scan chain。............

Question 22. What does the S22 violation mean and does this explain why 2 Sequential Cells are reported as synchronization elements in the **Sequential Cells Without Violations** summary?

[redacted]

[redacted] S22简而言之就是当一条chain里面有多个clock，或者多个clock edge的时候，如果缺了lock up latch，就会报。在lab7，有一个s22，报在了chain1上

```
---------------------------------------------------------------------------
DRC Violations which can affect ATPG Coverage
---------------------------------------------------------------------------

 Warning: Clock sdr_clk is connected to primary output sd_CK. (C17-1)
 Warning: Multiple clocks (pclk sys_clk) were used to shift scan chain chain1. (S22-1)
 Warning: Enable of bidi bus driver pad[10] connected to scan cell gate (I_ORCA_TOP/I_PCI_
9-1)
Information: There are 43 other cells with the same violation. (TEST-171)
```

所以一般看到S22的warn，我们得分析，是否缺了lock up latch

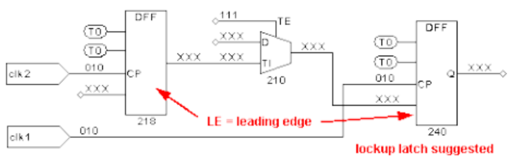
- Category K - LBIST Rules
- Category N - Netlist Rules
- Category P - Path Delay Rules
- Category R - Compressor Rules
- Category S - Scan Chain Rules
  - DRC Rule S1
  - DRC Rule S2
  - DRC Rule S3
  - DRC Rule S4
  - DRC Rule S5
  - DRC Rule S6
  - DRC Rule S7
  - DRC Rule S8
  - DRC Rule S9
  - DRC Rule S10
  - DRC Rule S11
  - DRC Rule S12
  - DRC Rule S13
  - DRC Rule S14
  - DRC Rule S15
  - DRC Rule S16
  - DRC Rule S17
  - DRC Rule S18
  - DRC Rule S19
  - DRC Rule S20
  - DRC Rule S21
  - DRC Rule S22
  - DRC Rule S23
  - DRC Rule S24
  - DRC Rule S25
  - DRC Rule S26
  - DRC Rule S27
  - DRC Rule S28
  - DRC Rule S29
  - DRC Rule S30
  - DRC Rule S32
- Category V - Vector Rules
- Category X - X-State Rules
- Category Y - PHDS Rules
- Category Z - Tristate Rules

Severity

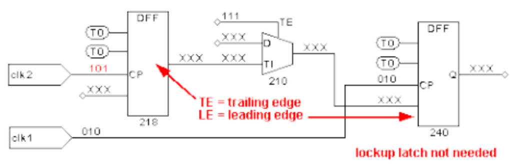
Warning

Examples

First example, LE to LE devices, LOCKUP latch suggested:



Second example, TE to LE devices, ignore S22 warning:



Description

All master gates of scan chain cells in the same scan chain should use a single shift clock. This rule is a safeguard to try to avoid a situation where chip tester clock skew results in tester pattern failures during the shifting of the scan chains. If a single clock is used then no clock skew due to the tester can occur. If more than one clock is used then any skew can result in the rejection at the tester of potentially good devices.

S2 and *C2* are the clock port names involved, and `chain_name` is the symbolic name of the scan chain as defined in the DRC procedure file used with the `run_drc` command.

This check is performed by inspecting the clocks sources of the state elements for all scan cells of a scan chain without regard to the polarity of the clocks or the phase relationship between them. A rule violation occurs when a scan chain uses more than one clock to shift data. A potential problem can exist, but you will need to further investigate the affected clock polarities. Only the first occurrence of an S22 violation on each scan chain is identified, so investigating one violation does not mean there are not others on the same scan chain. You can specify the `set_rules s22 -verbose` command to print all instances and clocks of the violating transition.

388 / 492 95.59%